Social Media Hate Speech and Toxicity Mining Using NLP

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Abstract - This paper covers the data collection, cleaning, and exploratory analysis work for our project on detecting hate speech and toxicity in social media using NLP. We used two datasets - the Google Civil Comments (Jigsaw) dataset with around 1.8M forum comments and the TweetEval Hate dataset with 9,000 tweets. We also tried the Reddit API but could not get access in time, so we document that decision and explain our alternatives. After cleaning, we kept 1,765,331 Jigsaw records and 8,954 Twitter records. These were combined into two views: an EDA dataset of 17,954 records and a modeling dataset of 58,954 records. Text was preprocessed using contraction expansion, URL removal, and stop-word filtering. TF-IDF with Truncated SVD reduced features from 50,000 to 200 dimensions. Our ten visualizations show around 91.7% class imbalance in Jigsaw, only 2.6% vocabulary overlap between the two datasets, and a natural two-cluster structure in the data.

CCS Concepts: • Information systems → Data mining; • Computing methodologies → NLP; • Applied computing → Natural language processing.

Keywords: hate speech, toxicity classification, NLP, TF-IDF, SVD, t-SNE, KMeans, TweetEval, Jigsaw Civil Comments, data mining

1 Introduction

Social media has grown a lot and with that, toxic and hateful content has become a real problem. Hate speech content that targets people based on race, religion, gender, or similar things is quite different from just someone being rude online. When a system treats both the same way, it ends up making wrong decisions about what to remove and what to leave up.

Our project looks at whether Transformer-based NLP models can do a better job than traditional ML methods at separating hate speech from general offensive language. If a model can get this right, platforms can respond differently to different kinds of content instead of applying one blanket policy.

This milestone is focused on data collection, cleaning, and understanding the data through visualizations. We tried to document every choice we made including the Reddit API attempt that did not work out so everything is reproducible and explainable.

Contributions: (1) A reproducible two-source pipeline via HuggingFace Datasets API; (2) documented Reddit API exploration and fallback justification; (3) full cleaning with before/after snapshots; (4) two merged dataset views (EDA: 17,954 records; modeling: 58,954 records); (5) text preprocessing pipeline; (6) TF-IDF + SVD feature engineering; (7) ten unique visualizations.

2 Background and Related Work

Some earlier work on hate speech detection split content into three categories: hate speech, offensive but not hateful, and neutral. The main finding was that classifiers tend to confuse the first two categories the most, which is exactly the problem we are trying to solve. Other studies on Twitter data have also shown that class imbalance is a major issue when non-toxic content is 90%+ of the data, models tend to just predict non-toxic for everything.

Pre-trained language models like BERT and DistilBERT have done well on hate speech detection tasks because they understand context rather than just counting words. However, models trained on one platform (like forum comments) often do not work well on another (like Twitter) because the writing style and vocabulary are so different. This domain gap is something we directly tried to account for in how we set up our data pipeline.

3 Data Acquisition

3.1 Selected Datasets

We picked two datasets that are widely used in NLP research and both available through the HuggingFace Datasets API using load_dataset(). Using an API also satisfies the project requirement for dynamic data collection.

Dataset	Source	Size / Key Fields
Google Civil Comments (Jigsaw)	google/civil_comments	1,804,874 rows; text + 7 toxicity scores
TweetEval Hate	cardiffnlp/tweet_eval (hate)	9,000 rows; text + binary label

Table 1: Primary dataset overview.

3.2 Reddit API - Attempted and Unavailable

We originally wanted to include Reddit as a third source. We set up a Reddit developer app and configured PRAW (Python Reddit API Wrapper) to collect posts from relevant subreddits. However, Reddit changed its API access rules in 2023, making it much harder for new apps to get free-tier access. We simply could not get it working in time.

We switched to HuggingFace sources for a few reasons: both datasets are still loaded via API calls so the dynamic access requirement is met; Jigsaw Civil Comments is a large, carefully labeled dataset from a real platform; TweetEval is the standard Twitter NLP benchmark; and together they cover both long forum comments and short tweets, which gives us the cross-platform variety we were hoping Reddit would add.

3.3 Additional Sources Considered

We also looked at HatExplain (has word-level annotations) but set it aside for now. YouTube Data API was tested but the 10,000 daily quota units were not enough since each search call costs 100 units. Mastodon API and Wikipedia EventStreams were also briefly checked but did not have

enough relevant hate speech content. Web scraping from public platforms is something we plan to explore in Milestone 3 to collect fresh, real-world data for testing our trained models.

4 Data Cleaning and Preprocessing

4.1 Initial Quality Assessment

When we first loaded both datasets, there were zero null values in either one, so no missing data handling was needed. Jigsaw started at 1,804,874 rows with 8 columns and Twitter had 9,000 rows with 2 columns. All data types were correct text as string, toxicity scores as float32, and Twitter label as int64.

4.2 Duplicate Removal

We removed exact duplicate rows from both datasets. Jigsaw had 17,955 duplicates (around 1%), leaving 1,786,919 rows. Twitter had 12 duplicates, leaving 8,988. Keeping duplicates would have made the model over-learn from repeated examples.

4.3 Short Text Filtering

We removed rows with fewer than 3 words since they do not have enough content for meaningful learning. From Jigsaw, 21,588 rows (1.21%) were dropped, leaving 1,765,331. From Twitter, 34 rows (0.38%) were dropped, leaving 8,954. Table 2 gives the complete before/after numbers.

Dataset	Raw	Dupes -	Short -	Clean
Jigsaw	1,804,874	17,955	21,588	1,765,331
Twitter	9,000	12	34	8,954

Table 2: Before/after cleaning summary.

4.6 Dataset Before & After

To visually confirm the effect of our cleaning steps, we captured raw and processed snapshots of each dataset. The raw snapshot shows the original data as loaded from HuggingFace before any deduplication or filtering. The processed snapshot shows the same view after all cleaning steps were applied.

	text	toxicity	severe_toxicity	obscene	threat	insult	identity_attack	sexual_explicit
0	This is so cool. It's like, 'would you want your mother to read this??' Really great idea, well done!	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	Thank you!! This would make my life a lot less anxiety-inducing. Keep it up, and don't let anyone get in your way!	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	This is such an urgent design problem; kudos to you for taking it on. Very impressive!	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	Is this something I'll be able to install on my site? When will you be releasing it?	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	[illegible]	0.894	0.021	0.000	0.000	0.872	0.021	0.000
5	[illegible]	0.667	0.048	0.638	0.000	0.333	0.000	0.010
6	[illegible]	0.458	0.051	0.305	0.000	0.254	0.000	0.220
7	[illegible]	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	The ranchers seem motivated by mostly by greed; no one should have the right to allow their animals destroy public land.	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	It was a great show. Not a combo I'd of expected to be good together but it was.	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Figure 14: Jigsaw - Raw Dataset (first few rows, before cleaning)

	text	toxicity	severe_toxicity	obscene	threat	insult	identity_attack	sexual_explicit	word_count
0	This is so cool. It's like, 'would you want your mother to read this??' Really great idea, well done!	0.000	0.000	0.000	0.000	0.000	0.000	0.000	19
1	Thank you!! This would make my life a lot less anxiety-inducing. Keep it up, and don't let anyone get in your way!	0.000	0.000	0.000	0.000	0.000	0.000	0.000	22
2	This is such an urgent design problem; kudos to you for taking it on. Very impressive!	0.000	0.000	0.000	0.000	0.000	0.000	0.000	16
3	Is this something I'll be able to install on my site? When will you be releasing it?	0.000	0.000	0.000	0.000	0.000	0.000	0.000	17
[illegible]	[illegible]	0.894	0.021	0.000	0.000	0.872	0.021	0.000	8
[illegible]	[illegible]	0.667	0.048	0.638	0.000	0.333	0.000	0.010	4
[illegible]	[illegible]	0.458	0.051	0.305	0.000	0.254	0.000	0.220	3
8	The ranchers seem motivated by mostly by greed; no one should have the right to allow their animals destroy public land.	0.000	0.000	0.000	0.000	0.000	0.000	0.000	21
9	It was a great show. Not a combo I'd of expected to be good together but it was.	0.000	0.000	0.000	0.000	0.000	0.000	0.000	18
10	Wow, that sounds great.	0.000	0.000	0.000	0.000	0.000	0.000	0.000	4

Figure 15: Jigsaw - Processed Dataset (after deduplication and filtering)

	text	label
0	@user nice new signage. Are you not concerned by Beatlemania -style hysterical crowds crongregating on you…	0
1	[illegible]	1
2	[illegible]	1
3	[illegible]	1
4	Hysterical woman like @user	0

Figure 16: Twitter - Raw Dataset (first few rows, before cleaning)

	text	label	word_count
0	@user nice new signage. Are you not concerned by Beatlemania -style hysterical crowds crongregating on you…	0	16
1	[illegible]	1	21
2	[illegible]	1	16
3	[illegible]	1	11
4	Hysterical woman like @user	0	4
5	Me flirting- So tell me about your father...	0	8
6	The Philippine Catholic bishops' work for migrant workers should focus on families who are "paying the great...	0	17
7	[illegible]	1	31
8	When cuffin season is finally over	0	6
9	Send home migrants not in need of protection, Peter Dutton tells UN, HEY DUTTON HOW ABOUT THE ONES THAT HAVE STAYED AND NOT LEFT THE COUNTRY WHEN THEY SHOULD OVERSTAYERS ? WHY DONT YOU GO AND ROUND ALL THEM UP ?	1	41

Figure 17: Twitter - Processed Dataset (after deduplication and filtering)

The snapshots confirm that text entries are clean UTF-8 strings, toxicity fields are correctly typed as float32 (Jigsaw) and int64 (Twitter), and no empty or malformed rows remain after processing.

4.4 Outlier Analysis

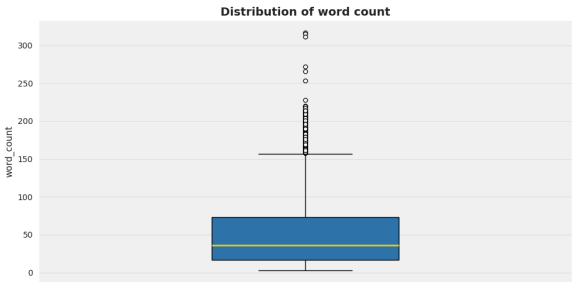

Figure 1: Jigsaw Word Count Boxplot

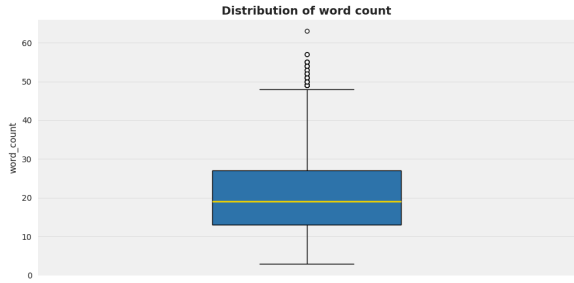

Figure 2: Twitter Word Count Boxplot

We checked for outliers in word count using IQR-based detection. For Jigsaw, 94,219 records were flagged as unusually long (upper bound = Q3 + 1.5×IQR). For Twitter, 118 records exceeded the upper bound of 48 words. We decided to keep all of them rather than removing them. Long comments exist in the real world and removing them would push our model to prefer shorter texts, which is not what we want. The two boxplots above (Figures 1 and 2) show the word count spread for each dataset separately. Jigsaw has a much wider spread with outliers going above 300 words, while Twitter is more tightly packed with most tweets under 50 words.

4.5 Data Quality Assessment

Completeness: Zero null values in both datasets after loading. No imputation required.

Consistency: `Jigsaw toxicity scores binarized at threshold 0.5 (label=1 if toxicity ≥ 0.5) to match Twitter's binary label scheme.`

Usability: Text columns are clean UTF-8 strings; numeric columns are float32 in [0,1]. The combined dataset is ready for both classical ML and Transformer fine-tuning.

5 Understanding the Data

5.1 Data Types and Schema

After cleaning, Jigsaw has 1,765,331 rows with one text column and seven float32 toxicity subscores. Twitter has 8,954 rows with one text column and one integer binary label. For combined analysis, we convert Jigsaw scores to a single binary label using the same 0.5 threshold.

5.2 Data Integration Strategy

Jigsaw is about 200 times larger than Twitter, so we could not just combine them directly. We built two separate merged views based on what each was needed for:

- EDA Dataset: 9,000 randomly sampled rows from Jigsaw (random_state=42) combined with all 8,954 Twitter rows = 17,954 total. Keeping sizes similar makes the cross-dataset comparison in visualizations fair.
- Modeling Dataset: 50,000 stratified rows from Jigsaw (25,000 non-toxic + 25,000 toxic) combined with all 8,954 Twitter rows = 58,954 total. Label split is 30,177 non-toxic (51.2%) and 28,777 toxic (48.8%) which is reasonably balanced.

We also added a source column to each row so we can track which dataset it came from during EDA. For Milestone 3, our plan is to first train on the 50,000 Jigsaw records and then fine-tune on Twitter to bridge the gap between the two domains.

5.3 Normality Check (Q-Q Plot)

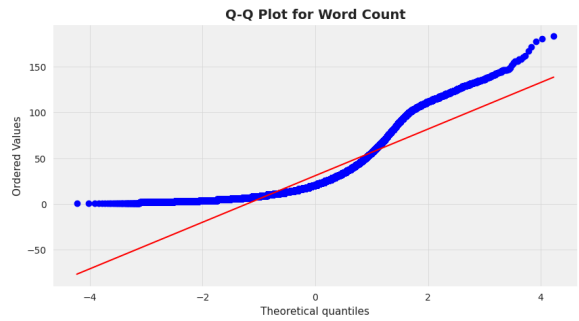

Figure 3: Q-Q Plot for Word Count

The Q-Q plot (Figure 3) confirms word count is right-skewed; most comments are short but a few are very long. This is fine since none of our planned models assume the input to be normally distributed.

6 Feature Engineering

6.1 Text Preprocessing Pipeline

Each text record went through a preprocessing pipeline before feature extraction. Classical ML models used the full pipeline below. For DistilBERT, we skipped lowercasing and some other steps since BERT's own tokenizer takes care of those.

Step	Description
Contraction Expansion	Resolve contractions (e.g. "don't" → "do not") via the contractions library + regex.
Lowercasing	Convert all characters to lowercase.
Mention Normalization	Replace @username patterns with token USER.
Hashtag Cleaning	Remove # symbol, retain the word.
URL Removal	Strip http/https and www patterns.
Tokenization	NLTK word_tokenize.
Stop-word Removal	Remove NLTK English stop words.

Table 3: Text preprocessing steps for classical ML models.

6.2 TF-IDF Vectorization

We used TF-IDF to convert text into numbers with these settings: ngram_range=(1,2) to capture both single words and word pairs, min_df=2 to ignore very rare terms, max_df=0.95 to ignore words that appear almost everywhere, and

max_features=50,000. The resulting sparse matrix is 58,954 × 50,000.

6.3 Dimensionality Reduction via Truncated SVD

We could not use standard PCA here because it needs the full dense matrix in memory, which is not feasible at 50,000 features. Instead we used Truncated SVD (also known as Latent Semantic Analysis), which is specifically designed to work on sparse matrices. This reduced our feature space from 50,000 to 200 dimensions. The final matrix is 58,954 × 200, which we then normalized using StandardScaler.

Method	Sparse Input	Memory	NLP Recommended
PCA	❌ No	❌ Poor	❌ No
Truncated SVD	✔ Yes	✔ Good	✔ Best

Table 4: PCA vs. Truncated SVD for sparse text features.

7 Exploratory Data Analysis

7.1 Class Distribution

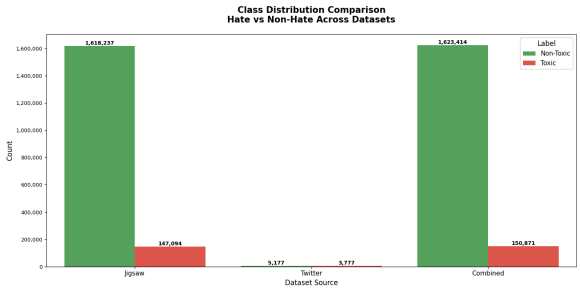

Figure 4: Class Distribution Bar Chart

The bar chart (Figure 4) makes the class imbalance obvious. In Jigsaw, 1,618,237 comments are non-toxic (91.7%) and only 147,094 are toxic (8.3%). Twitter is more balanced with around 5,177 non-hate and 3,777 hate samples. The combined EDA dataset sits at 74.8% non-toxic and 25.2% toxic overall. This imbalance needs to be handled carefully in modeling. We plan to use class weighting so the model does not just predict non-toxic for everything.

7.2 Text Length by Dataset

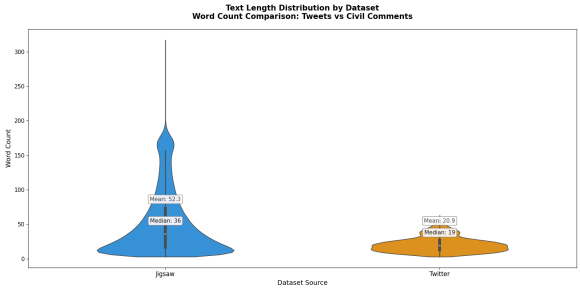

Figure 5: Text Length Violin Plot

The violin plot (Figure 5) shows Jigsaw comments are much longer on average - median 36 words, mean 52 words, and some going well above 100. Twitter is much shorter: median 19 words, mean around 21. This difference in text length between the two datasets means they represent quite different writing styles, which is one reason transferring a model from one to the other is not straightforward.

7.3 Text Length by Label

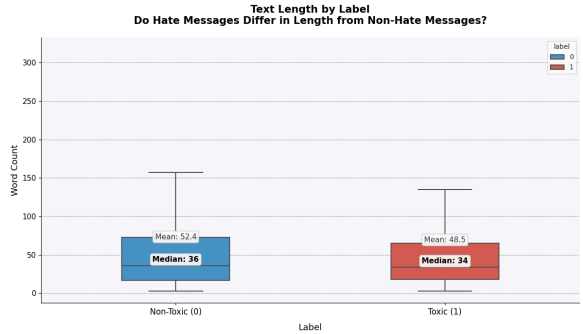

Figure 6: Text Length by Label Boxplot

Figure 6 shows that non-toxic and toxic texts have nearly identical word count distributions - medians of 36 and 34 words respectively. This tells us word count alone is not a useful feature for detecting toxicity. The actual content of the text matters much more.

7.4 Toxicity Score Distribution

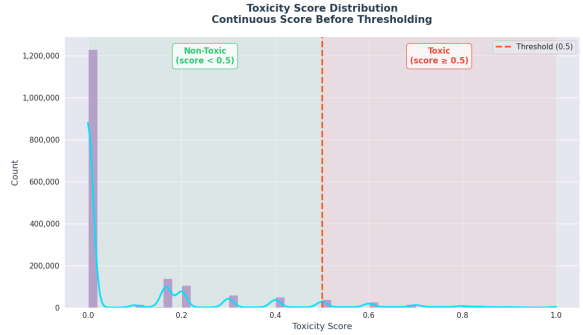

Figure 7: Toxicity Score Histogram

The histogram (Figure 7) shows most Jigsaw comments have toxicity scores very close to 0, with only a small number scoring near 1. Scores in the middle range (0.2 to 0.8) are quite rare. This bimodal shape confirms that using 0.5 as the threshold for binarization is reasonable and does not throw away much useful information.

7.5 Toxicity Feature Correlation

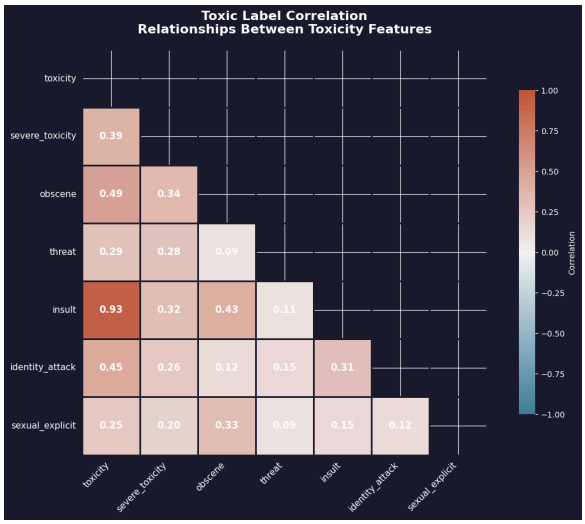

Figure 8: Toxicity Feature Correlation Heatmap

From the heatmap (Figure 8), toxicity and insult have a very strong correlation (r = 0.93), meaning they almost always appear together. Toxicity also has moderate correlation with obscene (r = 0.49) and identity_attack (r = 0.45). Interestingly, threat is quite independent from everything else (r = 0.09 to 0.29), which suggests that threatening language tends to show up on its own without other forms of toxicity.

7.6 Top Unigrams in Hate Content

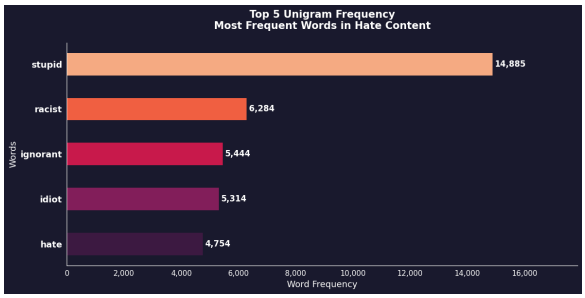

Figure 9: Top 5 Unigrams in Hate Content

The bar chart (Figure 9) shows the top 5 words in hate-labeled text. "Stupid" comes first with 14,885 occurrences, followed by "racist" (6,284), "ignorant" (5,444), "idiot" (5,314), and "hate" (4,754). Most of what gets labeled as toxic in this dataset is direct personal insults rather than organized group-based hate speech. This is an important distinction for our model to learn.

7.7 Word Cloud Comparison

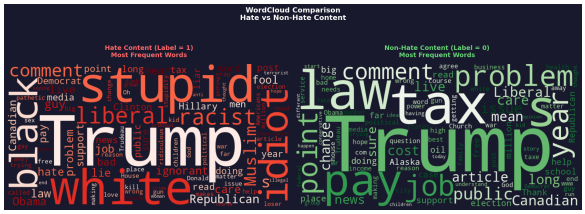

Figure 10: Word Cloud Comparison

The side-by-side word clouds (Figure 10) give a clear visual picture of how different the vocabulary is. In the hate content cloud, words like "stupid", "idiot", "racist", "Trump", "Muslim", "liberal", "black", "white", and "kill" stand out. The non-hate cloud has much more neutral words like "law", "tax", "job", "public", "comment". The vocabulary difference is quite stark and confirms that even simple bag-of-words features should give our models a reasonable signal to work with.

7.8 Vocabulary Overlap: Jigsaw vs. Twitter

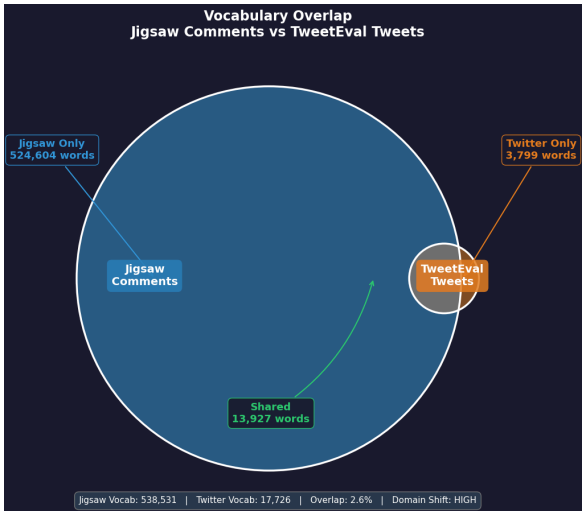

Figure 11: Vocabulary Overlap Venn Diagram

The Venn diagram (Figure 11) shows just how little overlap there is between the two datasets vocabulary-wise. Jigsaw has 538,531 unique tokens and Twitter has 17,726, but only 13,927 words are shared - just 2.6% overlap. Twitter has 3,799 words not found in Jigsaw at all, mostly slang, hashtags, and short forms. This large vocabulary gap is why a model trained on Jigsaw alone will likely struggle on Twitter data, and it is the main reason we plan to do domain-specific fine-tuning in Milestone 3.

7.9 t-SNE Cluster Structure

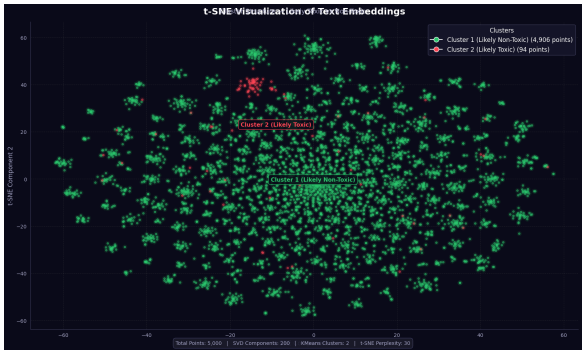

Figure 12: t-SNE Text Embeddings

The t-SNE plot (Figure 12) of 5,000 sampled points shows one large cluster (4,906 points, mostly non-toxic) and a smaller separate cluster of 94 points (likely toxic). The two clusters are not cleanly separated by a straight line, which means a simple linear classifier will probably not work very well here. This supports using non-linear models like LightGBM or DistilBERT.

7.10 KMeans Cluster Analysis

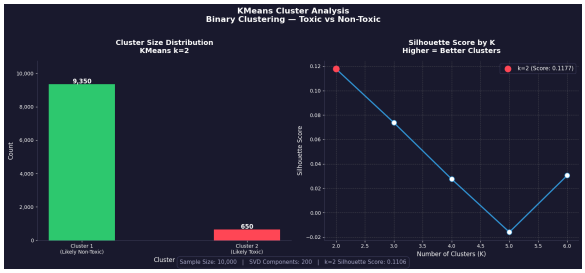

Figure 13: KMeans Cluster Analysis

KMeans with k=2 on 10,000 sampled records gives a silhouette score of 0.1177, which is the best score across k=2 to k=6. Cluster 1 has 9,350 points (93.5%) and Cluster 2 has 650 points (6.5%). These numbers are very close to the actual class split in the data, which shows the toxic and non-toxic content really does form two natural groups.

8 Ethics, Bias, and Limitations

Annotation Subjectivity: Jigsaw toxicity scores are crowd-sourced from human raters whose demographics and cultural backgrounds vary. Binarizing at 0.5 discards annotator uncertainty.

Demographic Bias: NLP toxicity models are known to disproportionately flag minority community language, including AAVE and in-group slang. This risk is present in our dataset and must be monitored in Milestone 3 by disaggregating metrics by identity-term presence.

Missing Context: Tweets are often replies to images, videos, or prior threads unavailable in text-only datasets. Sarcasm and irony may be miscategorized without this context.

Temporal Generalization: Both datasets reflect specific time periods. Hate speech language evolves; models may not generalize to current patterns without periodic retraining.

9 Modeling Plan

For Milestone 3, our plan is to train four classical ML models - Logistic Regression, Naive Bayes, Linear SVM, and LightGBM - using the TF-IDF + SVD feature matrix with class weighting to handle the imbalance. For deep learning, we will use DistilBERT in two stages: first pre-train on 50,000 Jigsaw records to learn general toxic language patterns, then fine-tune on all 8,954 Twitter records to adapt to social media writing style. We will evaluate all models on precision, recall, F1-score, and AUC-ROC, with a focus on recall for the toxic class. In addition, we plan to collect fresh social media data through web scraping to use as a real-world test set, which will help us see how well the models generalize beyond the labeled benchmark datasets.

10 Conclusion

This milestone covered everything from collecting raw data to getting it ready for model training. We collected two well-known NLP datasets, cleaned them up, merged them into usable formats, and explored them through ten visualizations. The main takeaways are: Jigsaw has very heavy class imbalance (91.7% non-toxic); only 2.6% of vocabulary is shared between Jigsaw and Twitter, showing a large domain gap; text length is not useful for predicting toxicity; most toxic content is personal insults; and the data naturally clusters into two groups matching the labels. We also documented why the Reddit API did not work and how our alternatives still meet the project requirements. The final 58,954-record dataset is ready for Milestone 3.

Acknowledgments

We want to thank our course instructors for their support during the milestone reviews. All datasets used in this project are publicly available and openly licensed, and were only used for academic purposes. Google Civil Comments is from Jigsaw/Google under CC0 license and TweetEval is from Cardiff NLP.

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